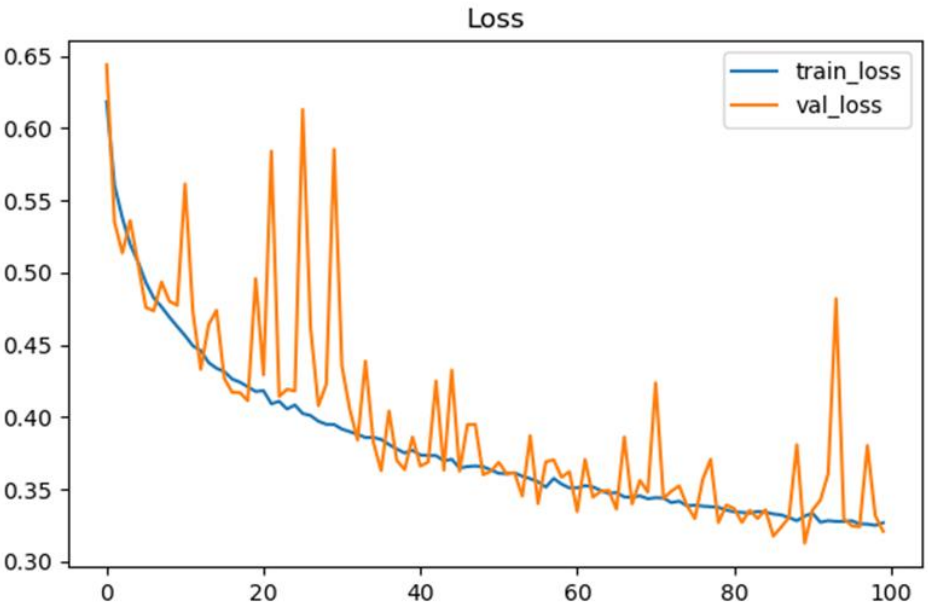
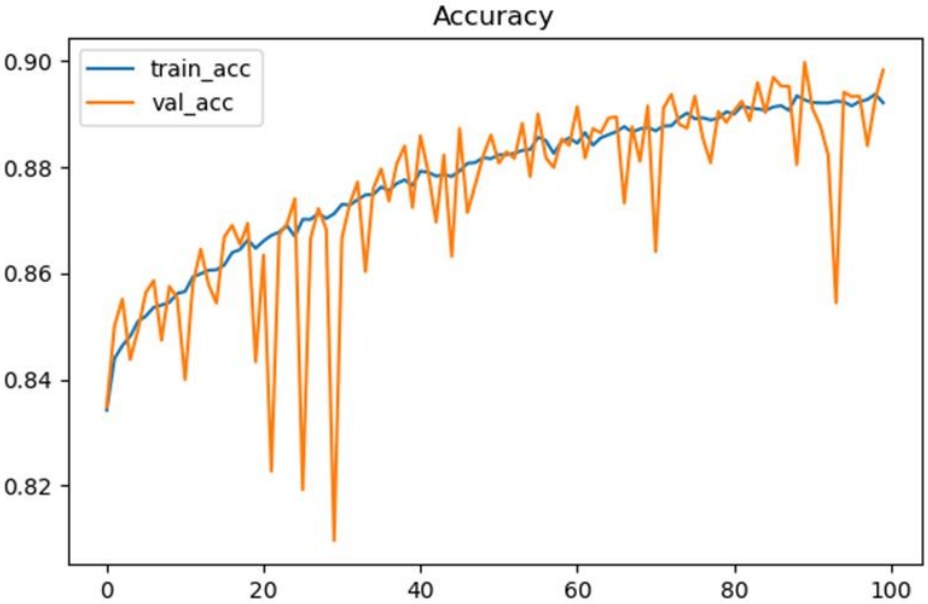
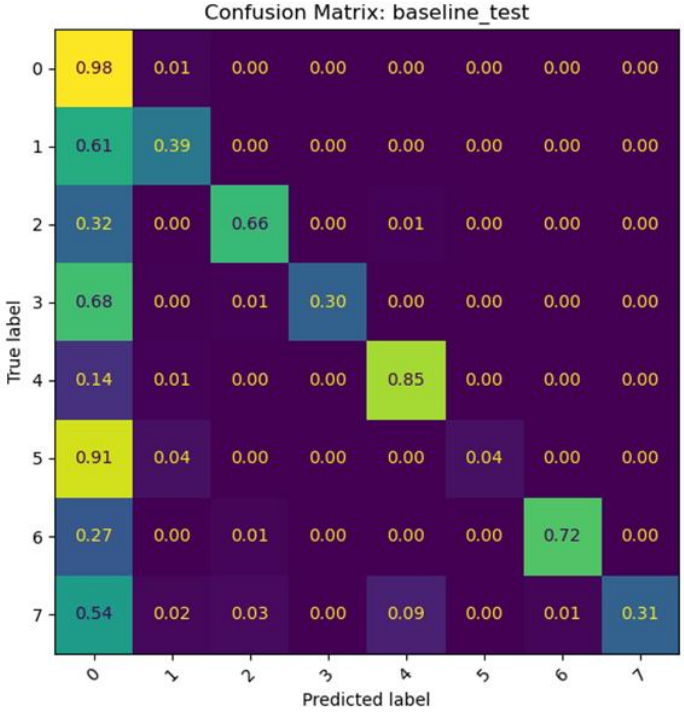
Supplementary data sheet

Supplementary Figure1: Training dynamics and Test set evaluation of the Baseline model

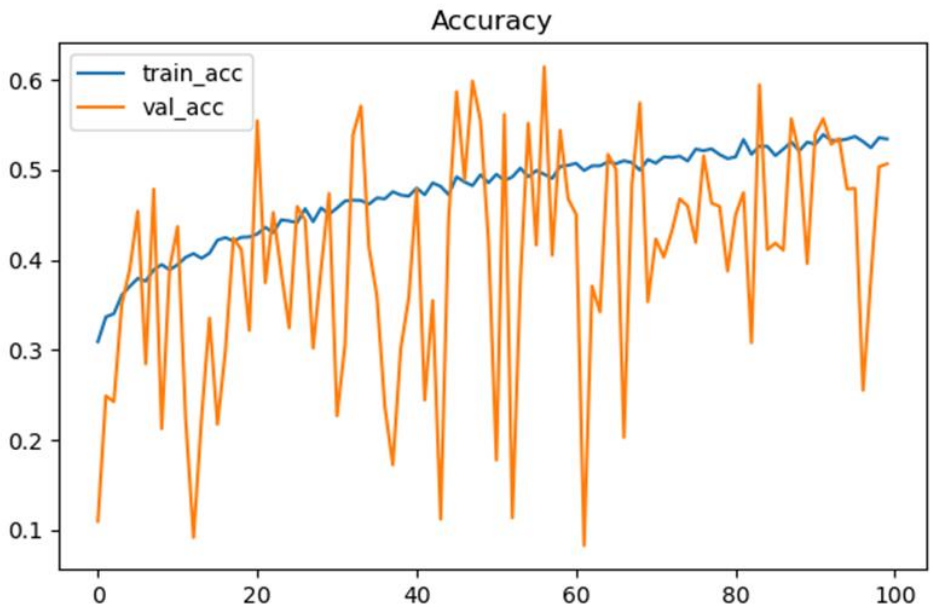


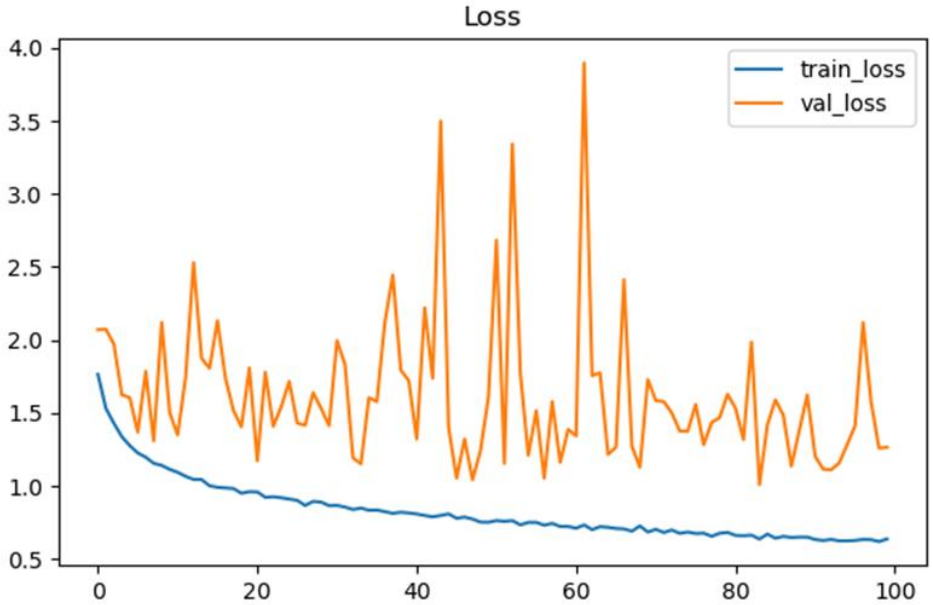


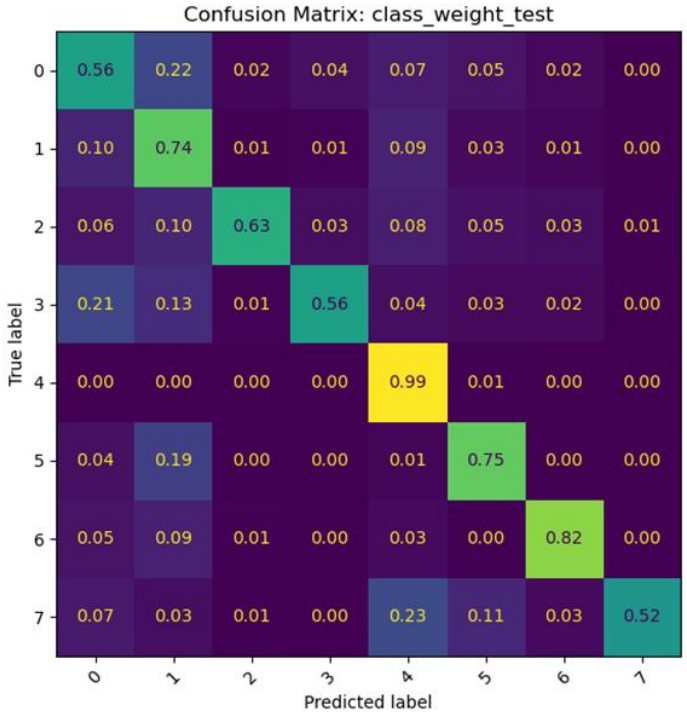



```
              precision    recall  f1-score   support

           0      0.909     0.980     0.943     18316
           1      0.681     0.387     0.493      1470
           2      0.794     0.665     0.724       343
           3      0.845     0.305     0.448       699
           4      0.937     0.848     0.890       525
           5      0.750     0.043     0.082        69
           6      0.898     0.715     0.796       344
           7      0.846     0.308     0.451       143

    accuracy                          0.899     21909
   macro avg      0.833     0.531     0.604     21909
weighted avg      0.890     0.899     0.884     21909

[baseline_test] macro_f1: 0.6035 | balanced_acc: 0.5313
Finished: baseline
```

Supplementary Figure2: Training dynamics and Test set evaluation of the Baseline model with inversed frequency based class reweighting



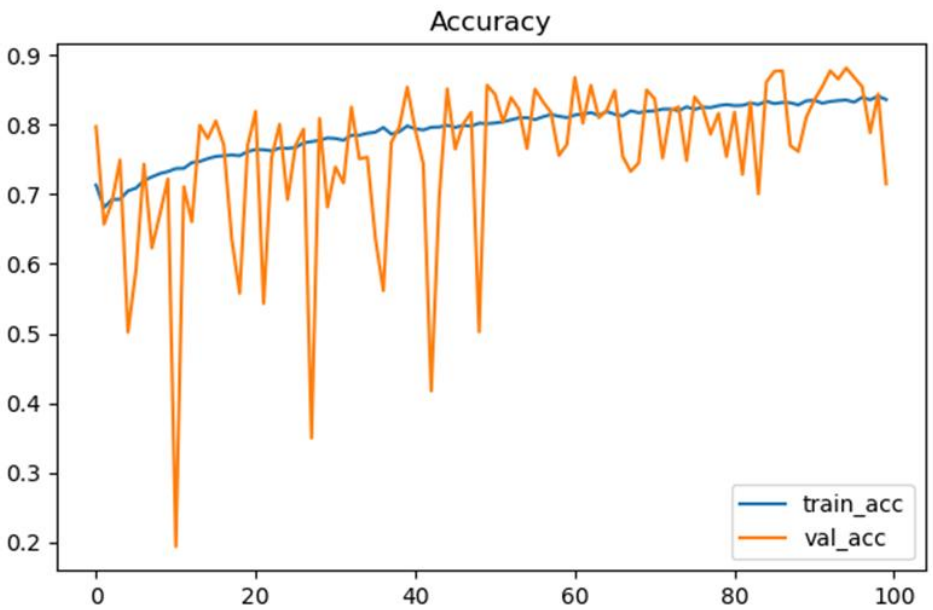


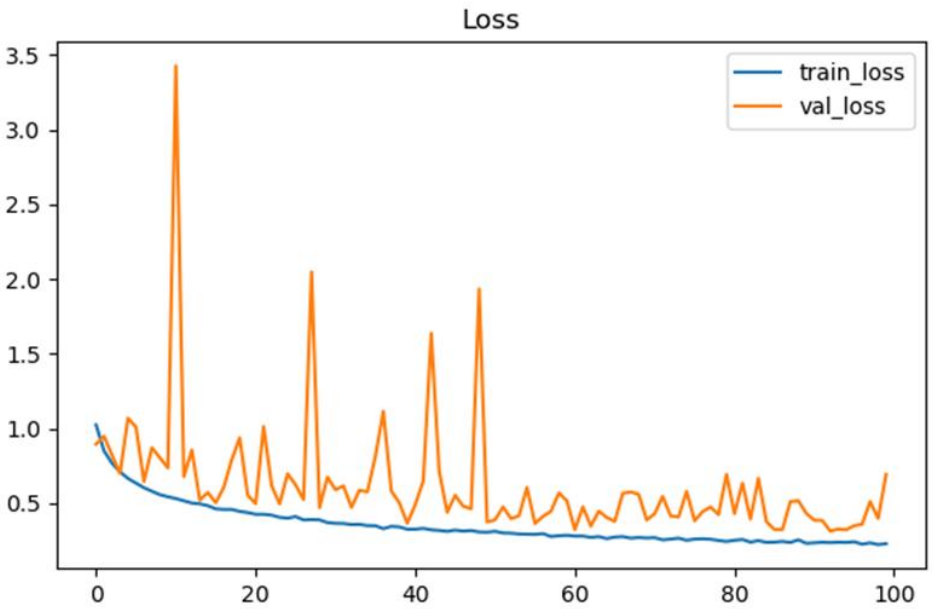


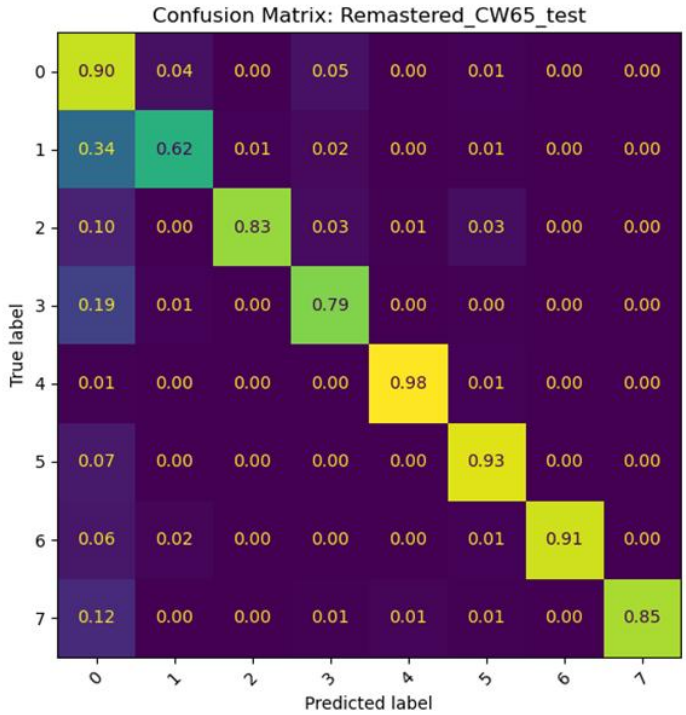
	precision	recall	f1-score	support
0	0.9672	0.5643	0.7128	18316
1	0.2089	0.7442	0.3262	1470
2	0.3839	0.6268	0.4762	343
3	0.3169	0.5622	0.4054	699
4	0.2441	0.9924	0.3919	525
5	0.0457	0.7536	0.0862	69
6	0.3648	0.8198	0.5049	344
7	0.5211	0.5175	0.5193	143
accuracy			0.5919	21909
macro avg	0.3816	0.6976	0.4279	21909
weighted avg	0.8539	0.5919	0.6591	21909

```
[class_weight_test] macro_f1: 0.4279 | balanced_acc: 0.6976
Finished: class_weight
```

Supplementary Figure3: Training dynamics and Test set evaluation of the Baseline model with soft(0.65) class reweighting



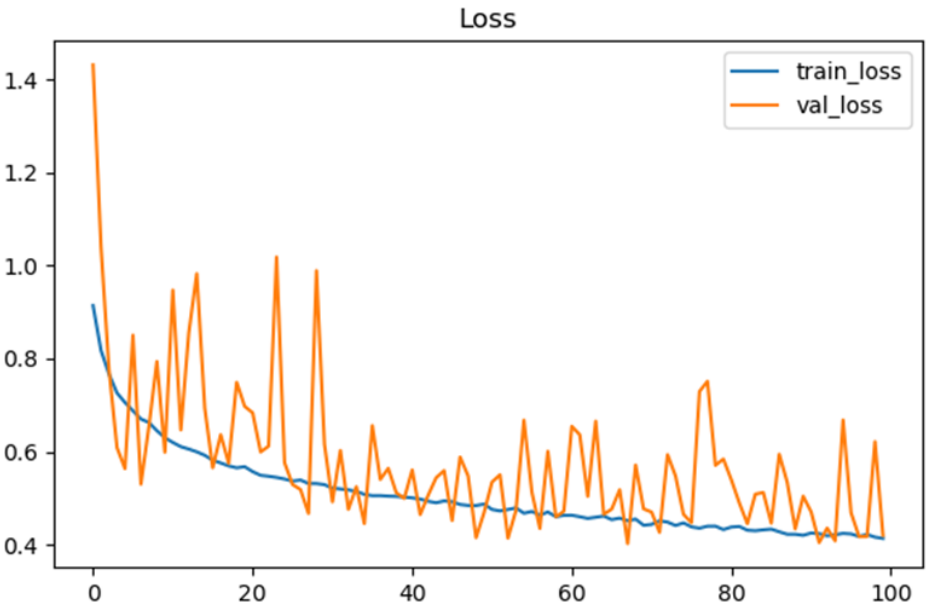


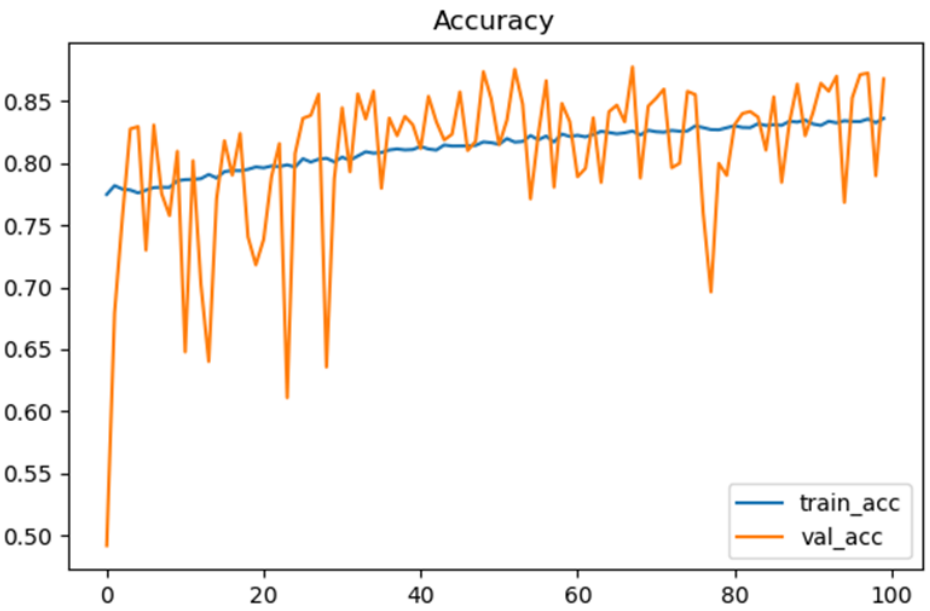


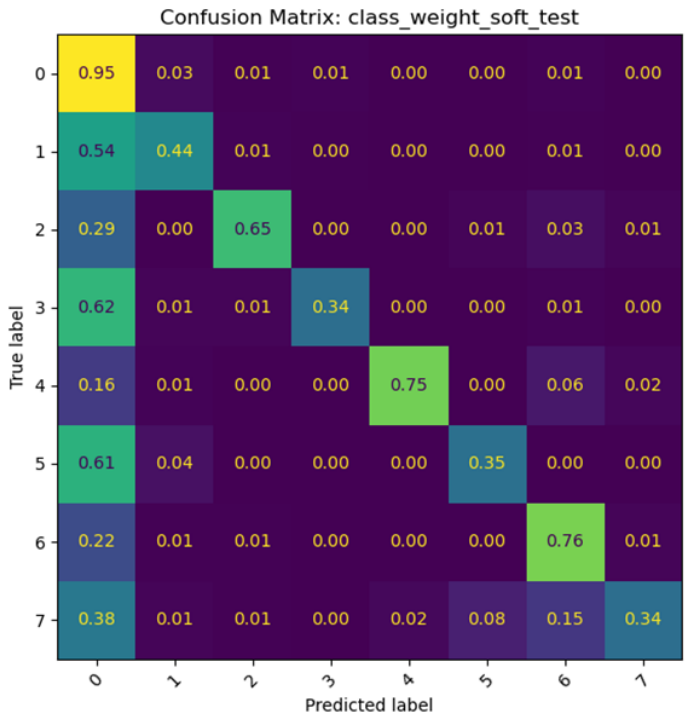
	precision	recall	f1-score	support
0	0.958	0.896	0.926	18316
1	0.530	0.620	0.572	1470
2	0.840	0.828	0.834	343
3	0.375	0.791	0.509	699
4	0.950	0.977	0.963	525
5	0.261	0.928	0.408	69
6	0.951	0.907	0.929	344
7	0.878	0.853	0.865	143
accuracy			0.875	21909
macro avg	0.718	0.850	0.751	21909
weighted avg	0.906	0.875	0.886	21909

[Remastered_CW65_test] macro_f1: 0.7506 | balanced_acc: 0.8499
Finished: Remastered_CW65

Supplementary Figure4: Training dynamics and Test set evaluation of the Baseline model with soft(0.50) class reweighting

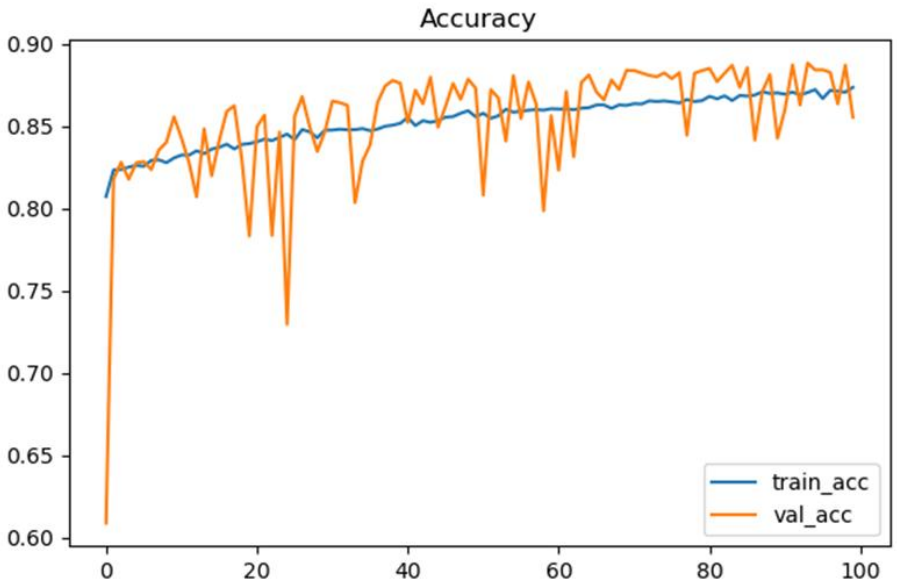


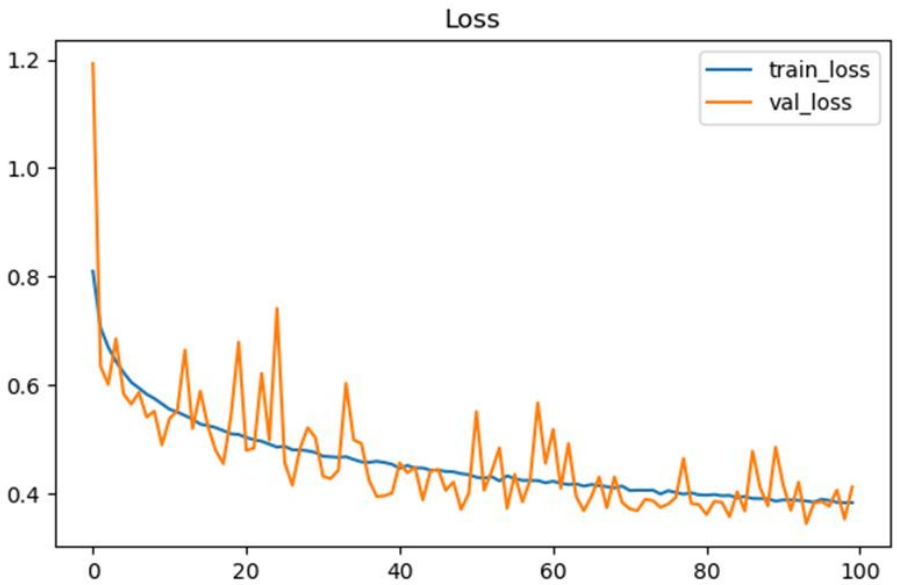


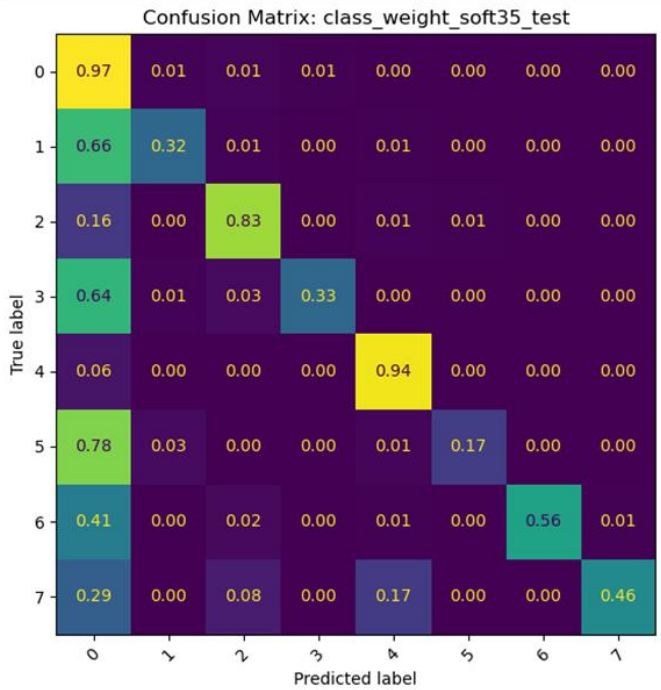



```
              precision    recall  f1-score   support

           0     0.9163    0.9465    0.9312     18316
           1     0.5663    0.4415    0.4962      1470
           2     0.6264    0.6501    0.6381       343
           3     0.5395    0.3419    0.4186       699
           4     0.9538    0.7467    0.8376       525
           5     0.3692    0.3478    0.3582        69
           6     0.5394    0.7558    0.6295       344
           7     0.5698    0.3427    0.4279       143

    accuracy                         0.8751     21909
   macro avg     0.6351    0.5716    0.5922     21909
weighted avg     0.8673    0.8751    0.8690     21909

[class_weight_soft_test] macro_f1: 0.5922 | balanced_acc: 0.5716
Finished: class_weight_soft
```

Supplementary Figure5: Training dynamics and Test set evaluation of the Baseline model with soft(0.35) class reweighting

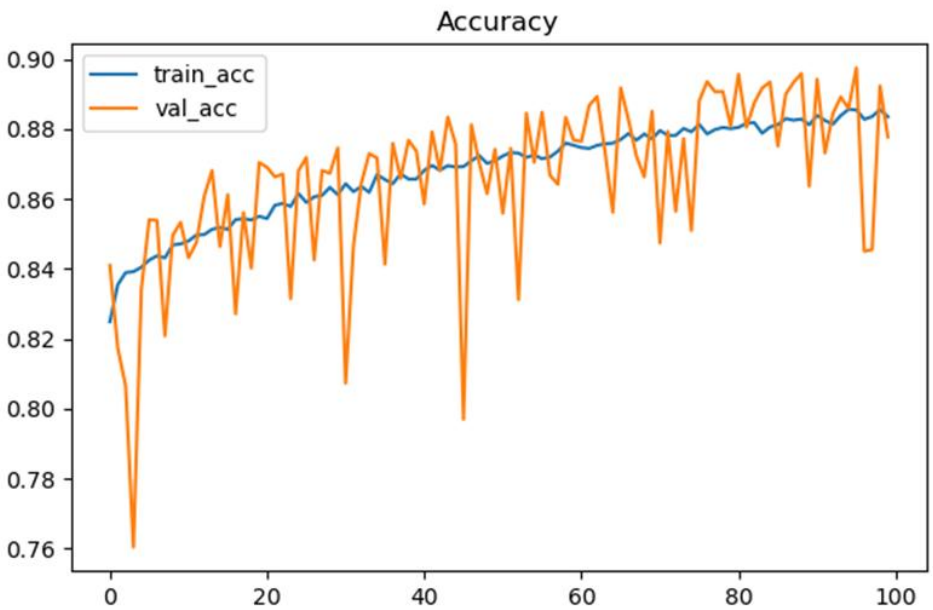


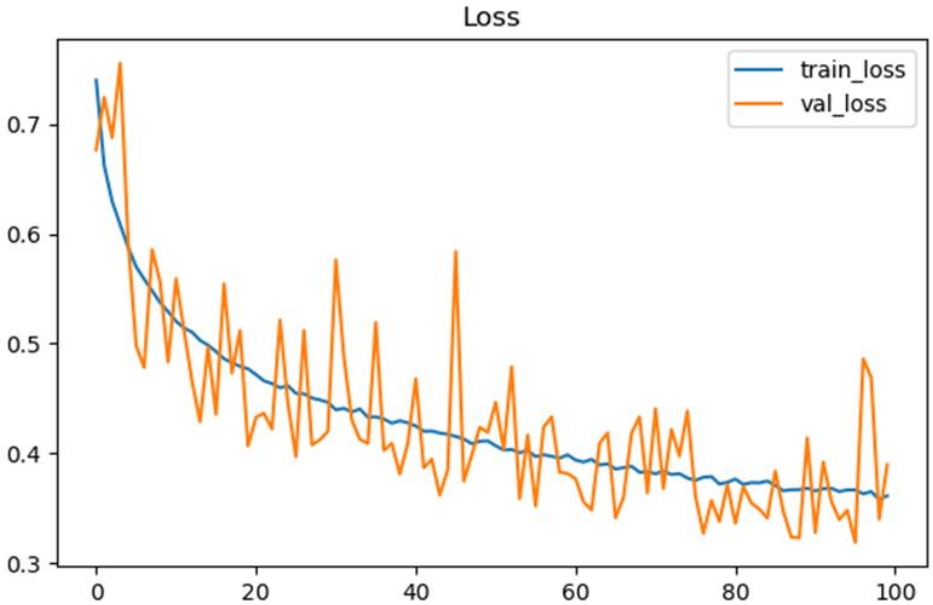


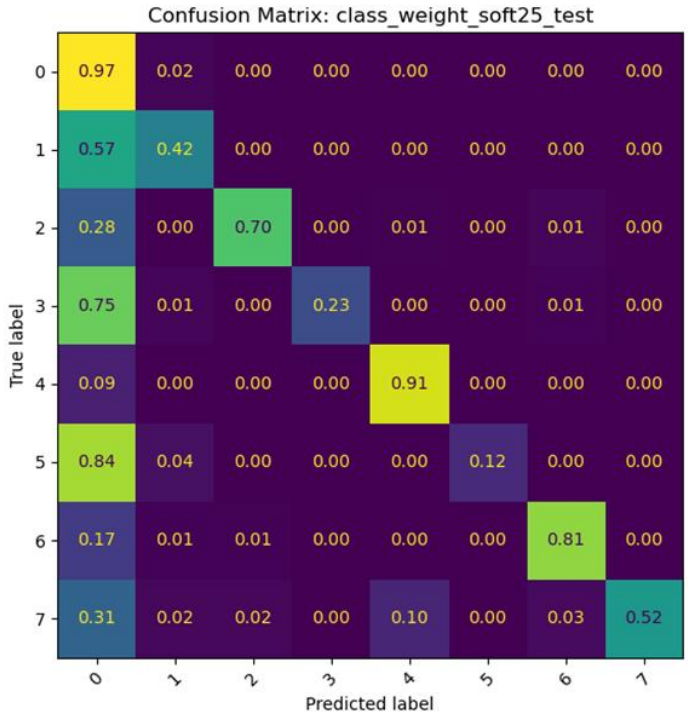



```
              precision    recall  f1-score   support

           0      0.911     0.966     0.938     18316
           1      0.721     0.322     0.446      1470
           2      0.484     0.825     0.610       343
           3      0.692     0.328     0.445       699
           4      0.829     0.945     0.883       525
           5      0.750     0.174     0.282        69
           6      0.923     0.555     0.693       344
           7      0.767     0.462     0.576       143

    accuracy                          0.888     21909
   macro avg      0.760     0.572     0.609     21909
weighted avg      0.881     0.888     0.874     21909

[class_weight_soft35_test] macro_f1: 0.6092 | balanced_acc: 0.5721
Finished: class_weight_soft35
```

Supplementary Figure6: Training dynamics and Test set evaluation of the Baseline model with soft(0.25) class reweighting

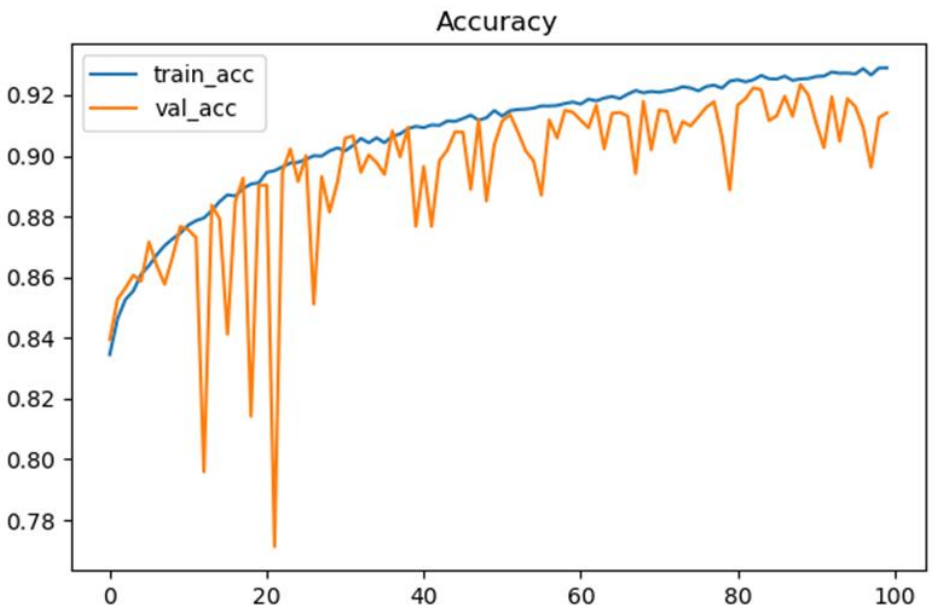


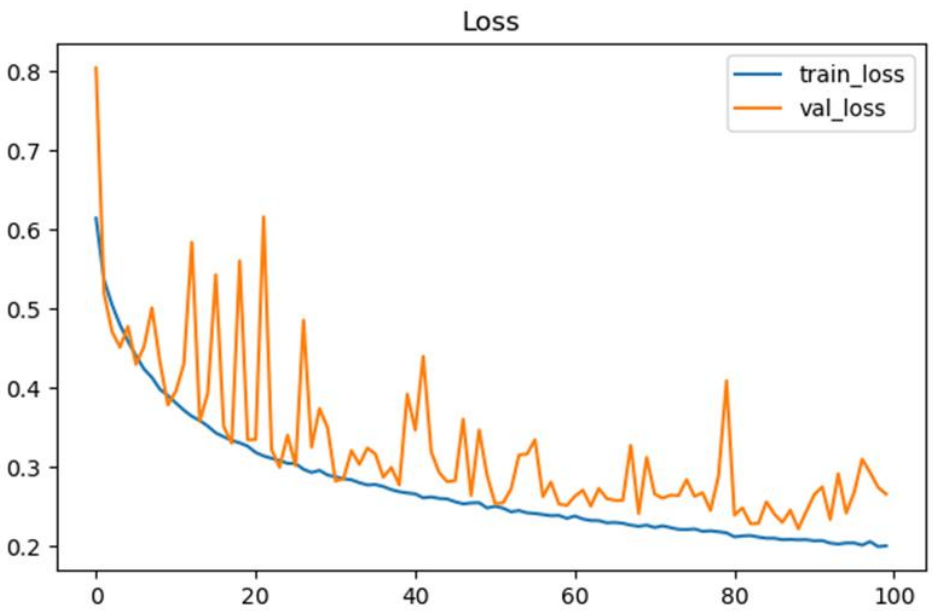


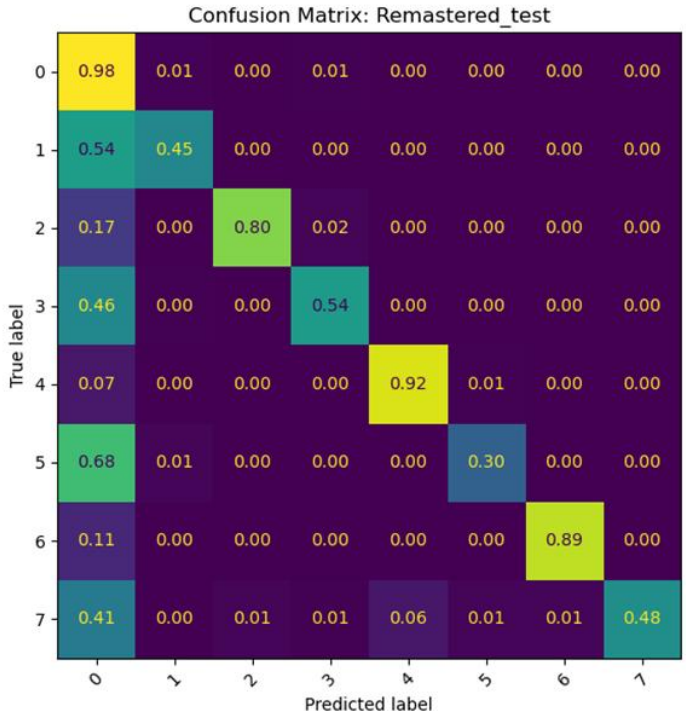



```
              precision    recall  f1-score   support

           0      0.914     0.971     0.942     18316
           1      0.635     0.423     0.508      1470
           2      0.755     0.703     0.728       343
           3      0.856     0.229     0.361       699
           4      0.921     0.910     0.916       525
           5      0.800     0.116     0.203        69
           6      0.809     0.811     0.810       344
           7      0.822     0.517     0.635       143

    accuracy                          0.897     21909
   macro avg      0.814     0.585     0.638     21909
weighted avg      0.889     0.897     0.884     21909

[class_weight_soft25_test] macro_f1: 0.6378 | balanced_acc: 0.5851
Finished: class_weight_soft25
```

Supplementary Figure7: Training dynamics and Test set evaluation of the Model2

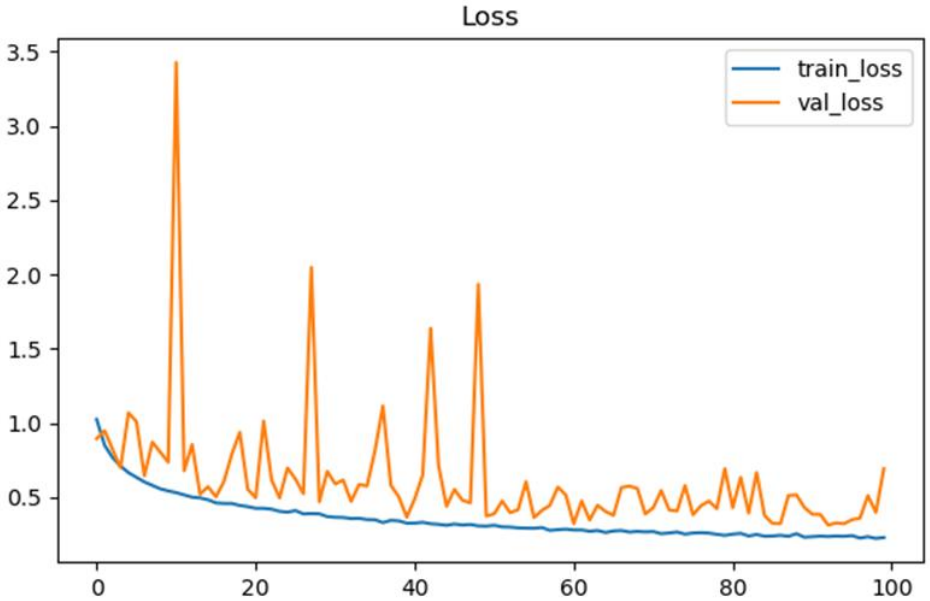


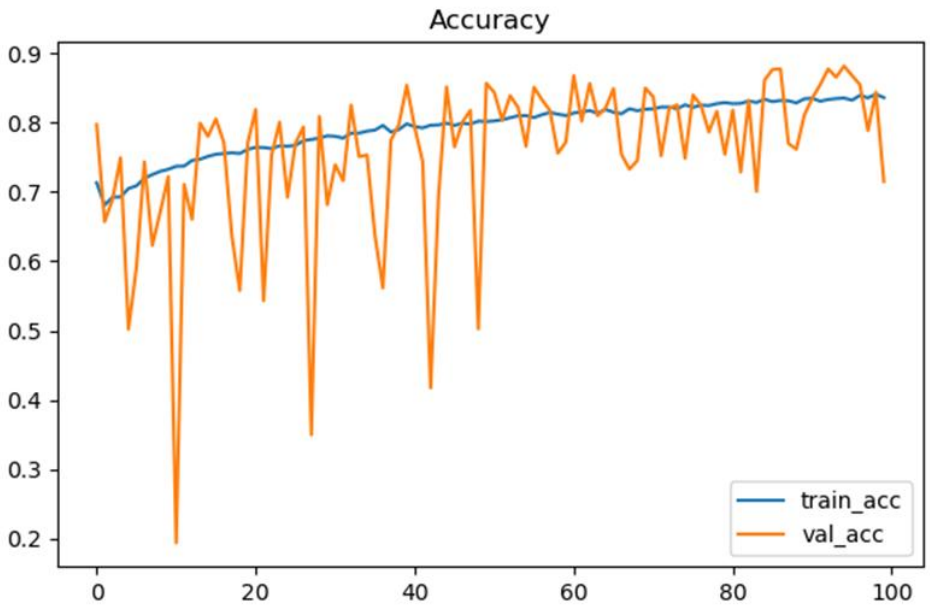


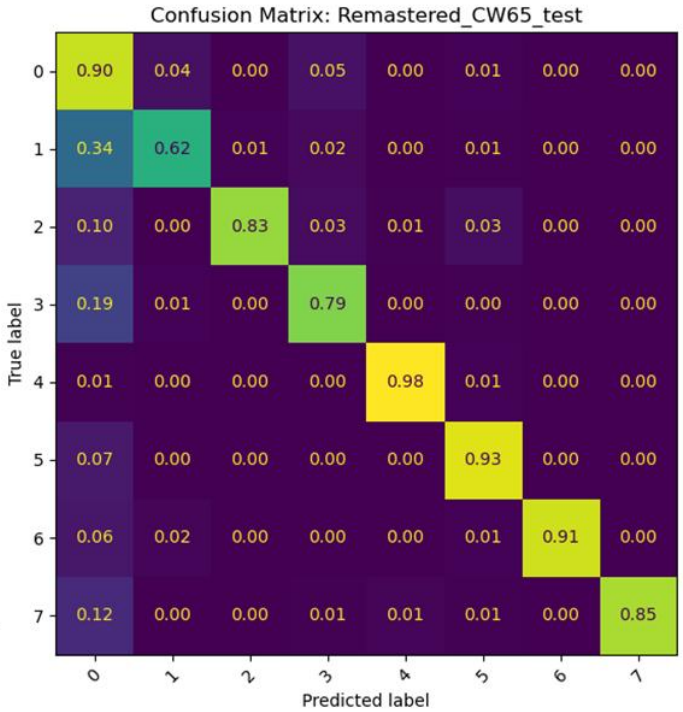



```
              precision    recall  f1-score   support

           0      0.930     0.980     0.954     18316
           1      0.761     0.453     0.568      1470
           2      0.935     0.799     0.862       343
           3      0.721     0.539     0.617       699
           4      0.972     0.924     0.947       525
           5      0.724     0.304     0.429        69
           6      0.984     0.887     0.933       344
           7      0.958     0.483     0.642       143

    accuracy                          0.920     21909
   macro avg      0.873     0.671     0.744     21909
weighted avg      0.913     0.920     0.912     21909

[Remastered_test] macro_f1: 0.7439 | balanced_acc: 0.6711
Finished: Remastered
```

Supplementary Figure8: Training dynamics and Test set evaluation of the Model2 with soft(0.65) class reweighting



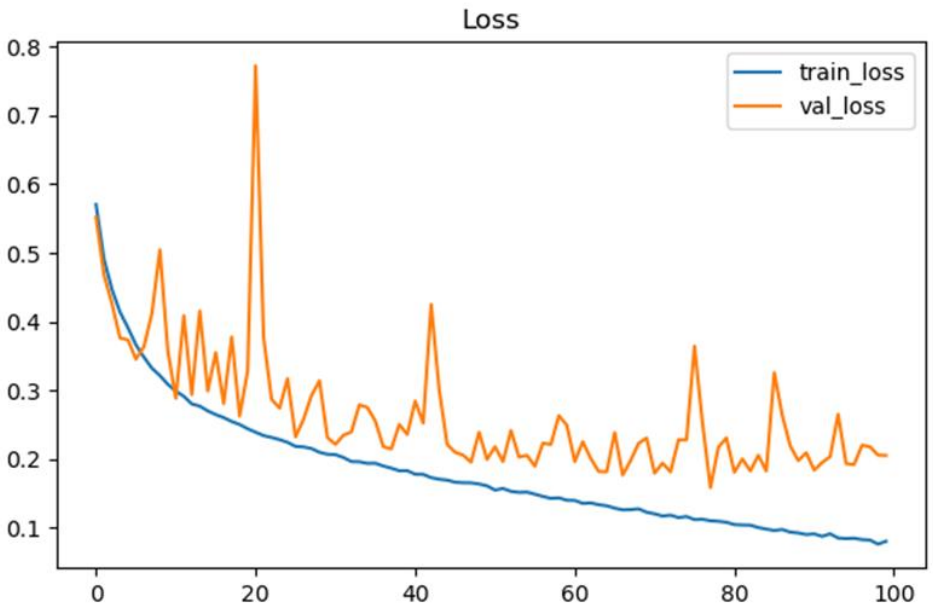


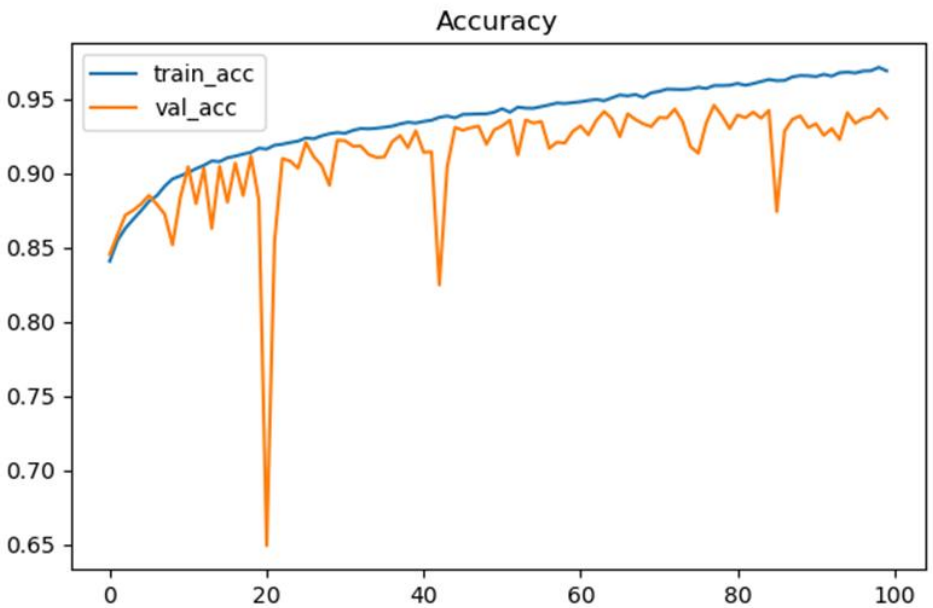


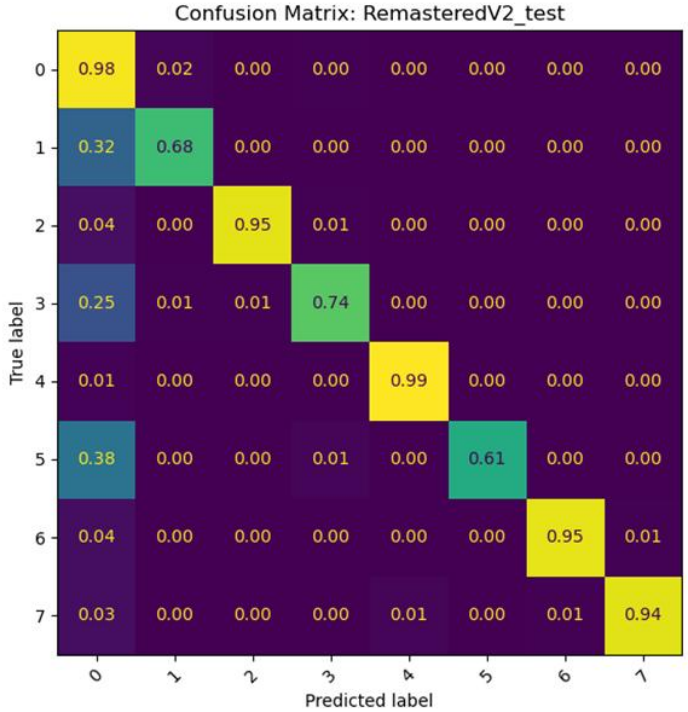
	precision	recall	f1-score	support
0	0.958	0.896	0.926	18316
1	0.530	0.620	0.572	1470
2	0.840	0.828	0.834	343
3	0.375	0.791	0.509	699
4	0.950	0.977	0.963	525
5	0.261	0.928	0.408	69
6	0.951	0.907	0.929	344
7	0.878	0.853	0.865	143
accuracy			0.875	21909
macro avg	0.718	0.850	0.751	21909
weighted avg	0.906	0.875	0.886	21909

```
[Remastered_CW65_test] macro_f1: 0.7506 | balanced_acc: 0.8499
Finished: Remastered_CW65
```

Supplementary Figure9: Training dynamics and Test set evaluation of the Model3

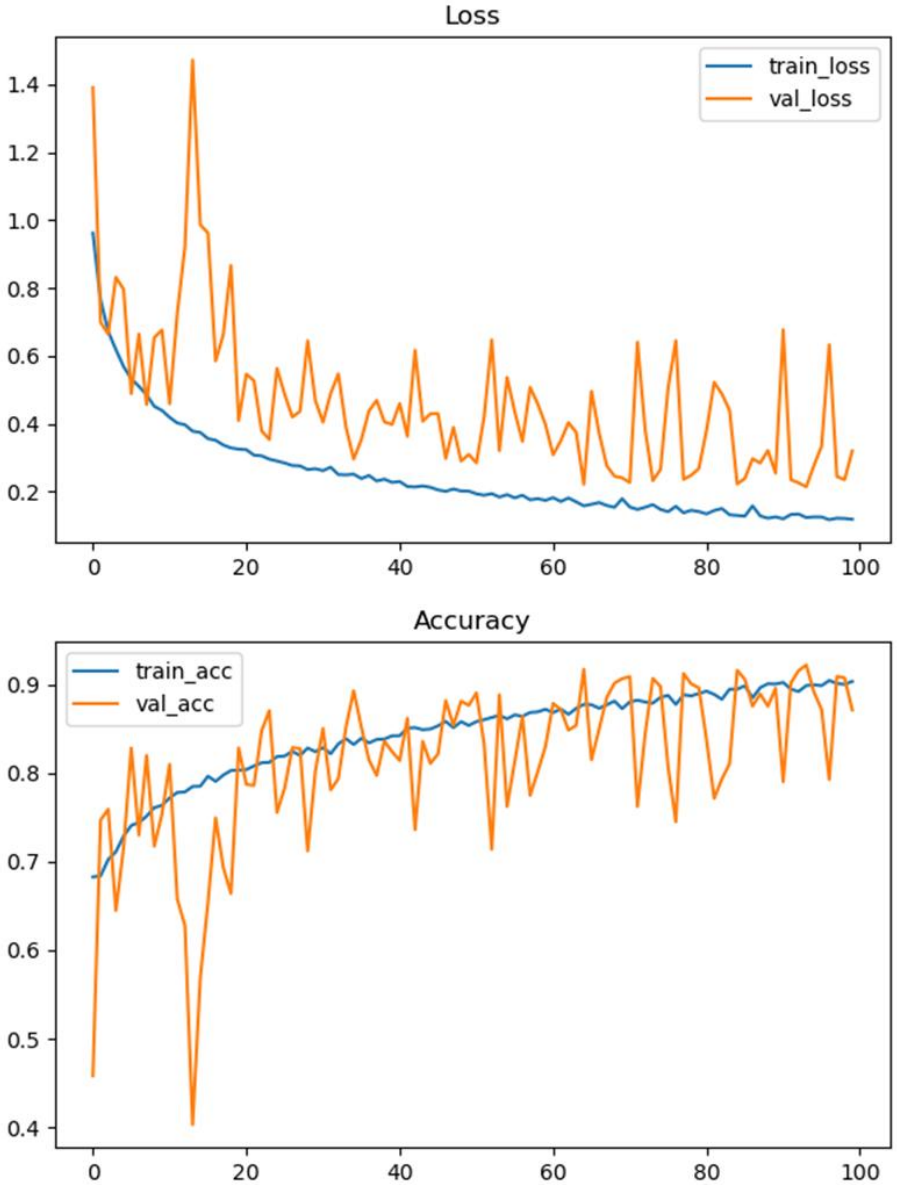


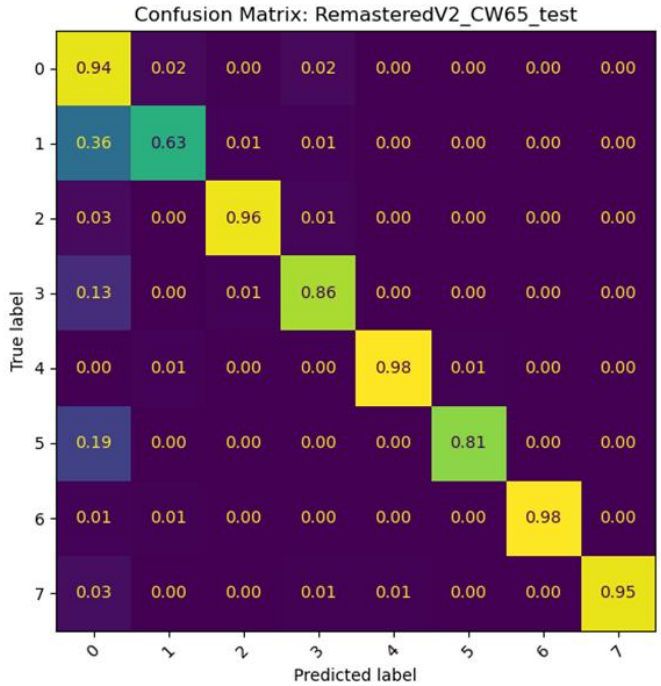





```
              precision    recall  f1-score   support

           0      0.962     0.976     0.969     18316
           1      0.764     0.680     0.720      1470
           2      0.881     0.948     0.913       343
           3      0.847     0.737     0.788       699
           4      0.994     0.989     0.991       525
           5      0.894     0.609     0.724        69
           6      0.988     0.953     0.970       344
           7      0.906     0.944     0.925       143

    accuracy                          0.946     21909
   macro avg      0.904     0.854     0.875     21909
weighted avg      0.944     0.946     0.945     21909

[RemasteredV2_test] macro_f1: 0.8750 | balanced_acc: 0.8544
Finished: RemasteredV2
```

Supplementary Figure10: Training dynamics and Test set evaluation of the Model3 with soft(0.65) class reweighting

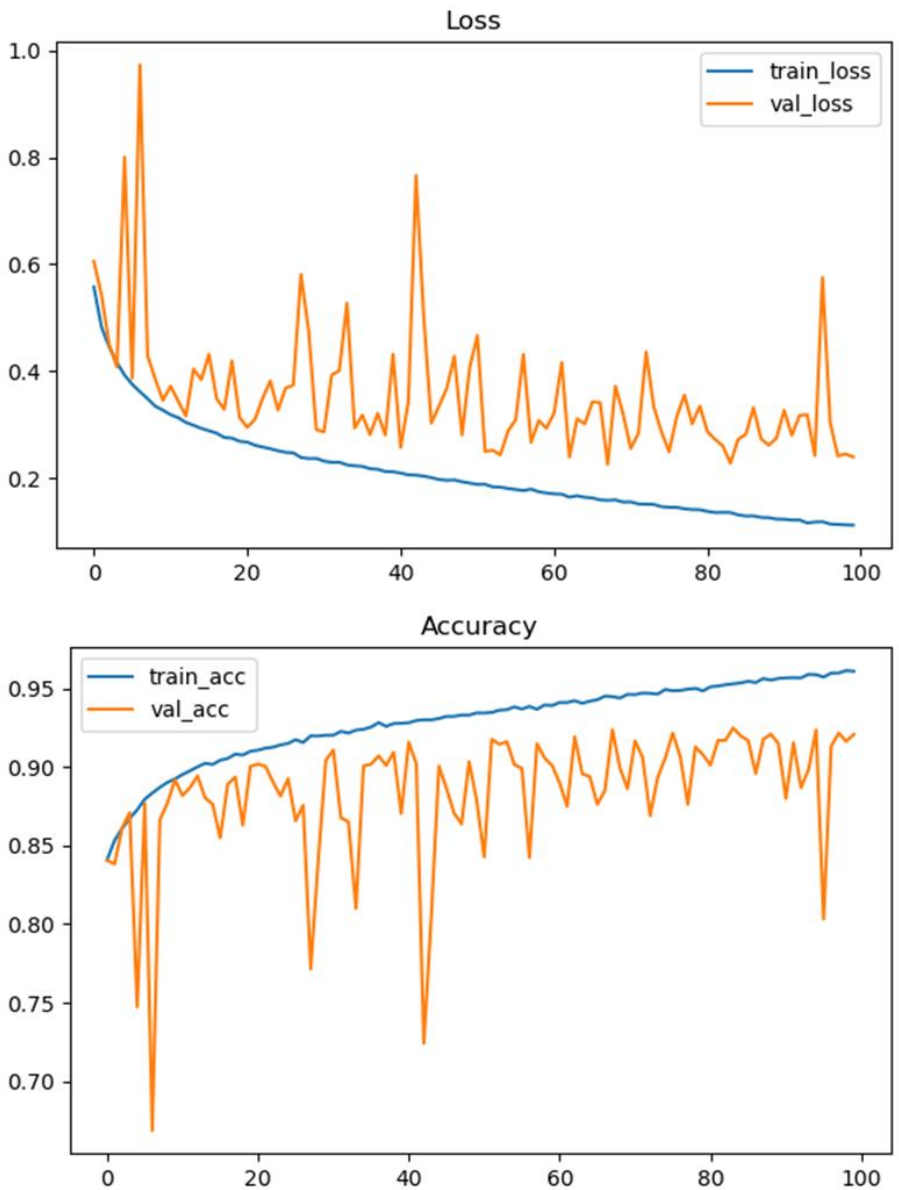


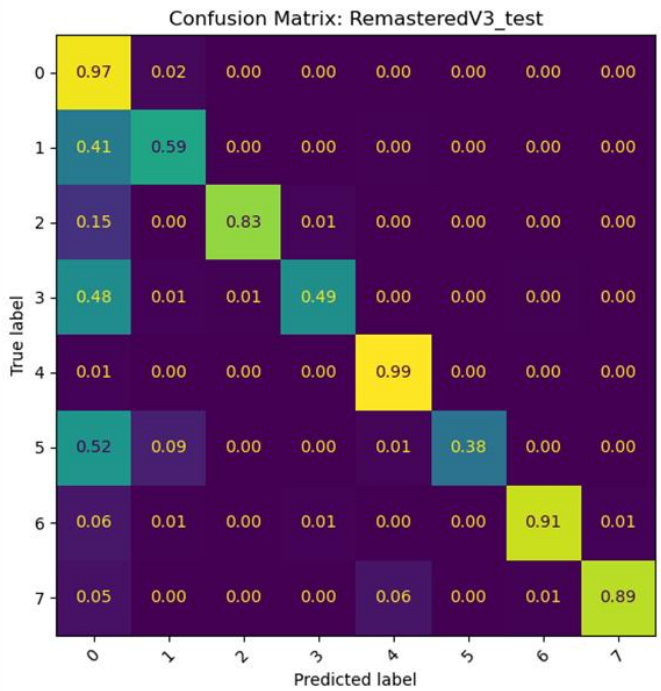


	precision	recall	f1-score	support
0	0.964	0.945	0.954	18316
1	0.671	0.625	0.647	1470
2	0.759	0.956	0.846	343
3	0.567	0.858	0.683	699
4	0.975	0.979	0.977	525
5	0.812	0.812	0.812	69
6	0.963	0.983	0.973	344
7	0.907	0.951	0.928	143
accuracy			0.922	21909
macro avg	0.827	0.889	0.853	21909
weighted avg	0.928	0.922	0.924	21909

```
[RemasteredV2_CW65_test] macro_f1: 0.8526 | balanced_acc: 0.8886
Finished: RemasteredV2_CW65
```

Supplementary Figure11: Training dynamics and Test set evaluation of the Model4

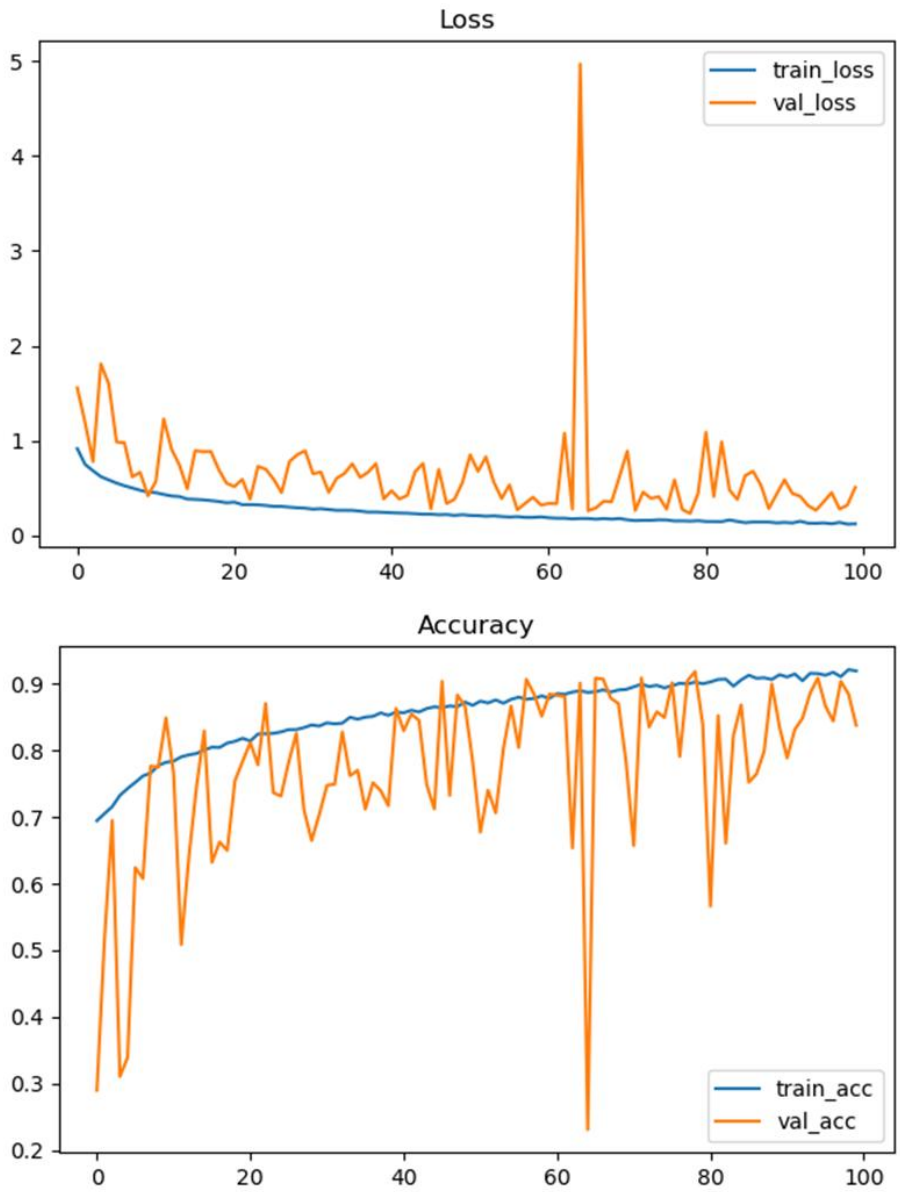


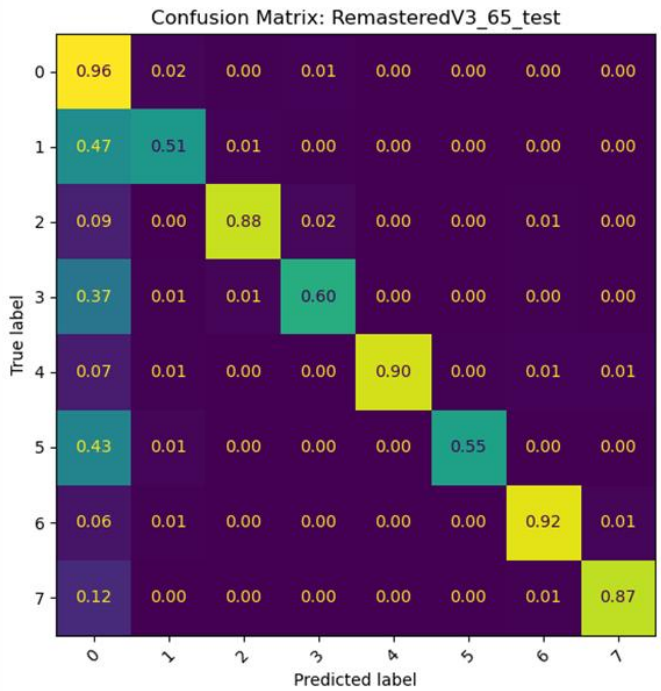


	precision	recall	f1-score	support
0	0.944	0.971	0.957	18316
1	0.691	0.587	0.635	1470
2	0.916	0.831	0.872	343
3	0.863	0.495	0.629	699
4	0.886	0.990	0.935	525
5	0.765	0.377	0.505	69
6	0.935	0.913	0.924	344
7	0.825	0.888	0.855	143
accuracy			0.925	21909
macro avg	0.853	0.757	0.789	21909
weighted avg	0.921	0.925	0.921	21909

```
[RemasteredV3_test] macro_f1: 0.7889 | balanced_acc: 0.7565
Finished: RemasteredV3
```

Supplementary Figure12: Training dynamics and Test set evaluation of the Model4 with soft(0.65) class reweighting

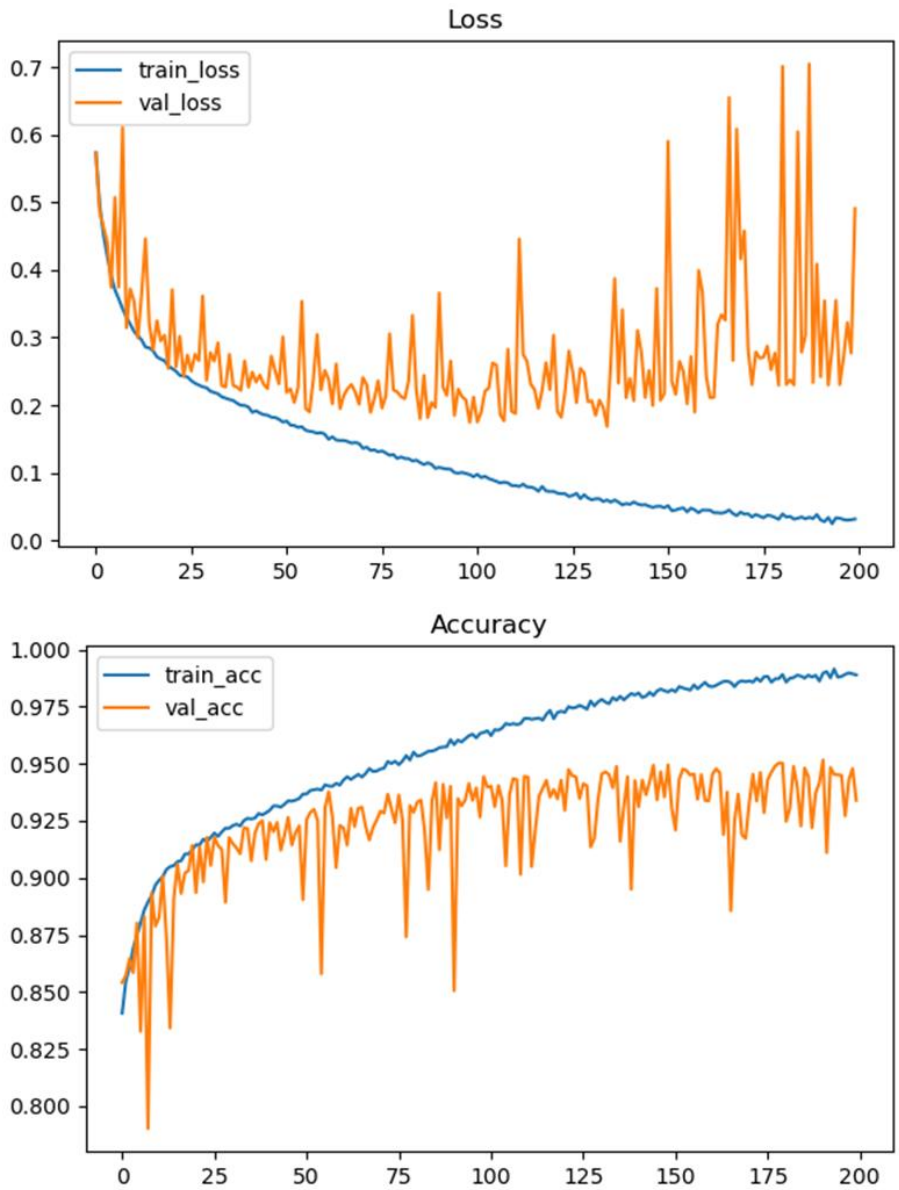


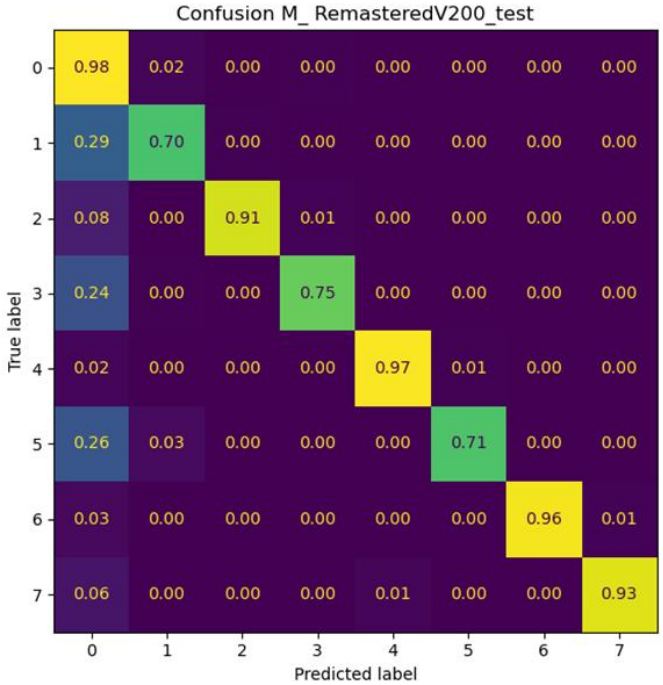


	precision	recall	f1-score	support
0	0.942	0.965	0.953	18316
1	0.705	0.513	0.594	1470
2	0.734	0.883	0.802	343
3	0.722	0.601	0.656	699
4	0.983	0.903	0.941	525
5	0.655	0.551	0.598	69
6	0.828	0.924	0.874	344
7	0.801	0.874	0.836	143
accuracy			0.918	21909
macro avg	0.796	0.777	0.782	21909
weighted avg	0.913	0.918	0.914	21909

```
[RemasteredV3_65_test] macro_f1: 0.7818 | balanced_acc: 0.7768
Finished: RemasteredV3_65
```

Supplementary Figure13: Training dynamics and Test set evaluation of the Model3 over 200 epochs





	precision	recall	f1-score	support
0	0.964	0.978	0.971	18316
1	0.769	0.699	0.732	1470
2	0.904	0.910	0.907	343
3	0.876	0.751	0.809	699
4	0.996	0.973	0.985	525
5	0.831	0.710	0.766	69
6	0.991	0.962	0.976	344
7	0.937	0.930	0.933	143
accuracy			0.949	21909
macro avg	0.908	0.864	0.885	21909
weighted avg	0.948	0.949	0.948	21909

[RemasteredV200_test] macro_f1: 0.8848 // balanced_acc: 0.8641